UNIT 4 : Lab Experiments

Slot C

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Course : Cryptography And Network Security **Course Code : CSA5104**

Program 1: IPSec Architecture Simulation

Aim

To design and implement a simulation demonstrating the IPSec architecture and the interaction between its components (SPD, SAD, AH, ESP, and IKE).

Algorithm

1. Start.
2. Define the core IPSec components: Security Policy Database (SPD), Security Association Database (SAD), AH module, ESP module and IKE module.
3. Create an outgoing IP packet with source, destination, protocol and payload.
4. Check the packet against the SPD to decide whether to bypass, discard or protect it.
5. If protection is required, look up the matching Security Association (SA) in the SAD.
6. Route the packet to the AH module, ESP module, or both, based on the SA parameters.
7. Apply the selected protocol(s) and forward the protected packet.
8. Display the interaction sequence between the components.
9. Stop.

Program Code (C++)

```
// Program 1: IPSec Architecture Simulation
#include <iostream>
#include <string>
#include <map>
using namespace std;

struct SPDEntry { string dest; string action; }; // action: BYPASS, DISCARD, PROTECT
struct SAEntry  { string protocol; string mode; string key; };

int main() {
   map<string, SPDEntry> spd;
   map<string, SAEntry> sad;

   spd["192.168.1.10"] = {"192.168.1.10", "PROTECT"};
   spd["192.168.1.20"] = {"192.168.1.20", "BYPASS"};

   sad["192.168.1.10"] = {"ESP", "TUNNEL", "K1A2B3C4"};

   string dest, payload;
   cout << "Enter destination IP : ";
   cin >> dest;
   cout << "Enter payload       : ";
   cin >> payload;

   cout << "\n--- IPSec Component Interaction ---\n";
   cout << "[1] Outbound packet created for " << dest << endl;
```

```
    if (spd.find(dest) == spd.end()) {
        cout << "[2] SPD: no matching policy -> DISCARD" << endl;
        return 0;
    }

    cout << "[2] SPD lookup -> action = " << spd[dest].action << endl;

    if (spd[dest].action == "BYPASS") {
        cout << "[3] Packet forwarded without IPSec processing." << endl;
    } else if (spd[dest].action == "DISCARD") {
        cout << "[3] Packet discarded by policy." << endl;
    } else {
        if (sad.find(dest) == sad.end()) {
            cout << "[3] SAD: no SA found -> trigger IKE negotiation." << endl;
        } else {
            SAEntry sa = sad[dest];
            cout << "[3] SAD lookup -> protocol = " << sa.protocol
                 << ", mode = " << sa.mode << endl;
            cout << "[4] " << sa.protocol << " module processes payload \""
                 << payload << "\" using key " << sa.key << endl;
            cout << "[5] Protected packet forwarded to " << dest << " in "
                 << sa.mode << " mode." << endl;
        }
    }
    return 0;
}
```

Input

Destination IP = 192.168.1.10
Payload = CONFIDENTIAL_REPORT

Output

```
Enter destination IP : 192.168.1.10
Enter payload        : CONFIDENTIAL_REPORT

--- IPSec Component Interaction ---
[1] Outbound packet created for 192.168.1.10
[2] SPD lookup -> action = PROTECT
[3] SAD lookup -> protocol = ESP, mode = TUNNEL
[4] ESP module processes payload "CONFIDENTIAL_REPORT" using key K1A2B3C4
[5] Protected packet forwarded to 192.168.1.10 in TUNNEL mode.
```

Result

The simulation successfully demonstrated the IPSec architecture, showing how a packet flows through the SPD, SAD, and the AH/ESP processing modules before transmission.

Program 2: Security Association (SA) Database and SA Lookup

Aim

To develop a program that creates a Security Association (SA) database and simulates SA lookup for incoming IP packets using the SPI, destination address and protocol.

Algorithm

1. Start.
2. Define an SA structure containing SPI, destination address, protocol, encryption algorithm and key.
3. Insert multiple SA entries into the SA database (SAD), keyed by (SPI, destination).
4. Read an incoming packet's SPI, destination address and protocol.
5. Search the SAD for a matching entry.
6. If found, display the SA parameters that will be used to process the packet.
7. If not found, report that the packet cannot be processed and must be dropped.
8. Stop.

Program Code (C++)

```
// Program 2: SA Database and SA Lookup
#include <iostream>
#include <string>
#include <vector>
using namespace std;

struct SA {
   unsigned int spi;
   string destIP;
   string protocol;
   string algorithm;
   string key;
};

SA* lookupSA(vector<SA>& db, unsigned int spi, const string& dest) {
   for (auto &sa : db)
      if (sa.spi == spi && sa.destIP == dest)
         return &sa;
   return nullptr;
}

int main() {
   vector<SA> sad = {
      {1001, "192.168.1.10", "ESP", "AES-128", "K1A2B3C4D5E6F708"},
      {1002, "192.168.1.20", "AH",  "HMAC-SHA1", "K9F8E7D6C5B4A302"},
      {1003, "192.168.1.30", "ESP", "3DES", "K2233445566778899"}
   };

   unsigned int spi;
   string dest;
   cout << "Enter incoming SPI         : ";
   cin >> spi;
   cout << "Enter incoming destination  : ";
   cin >> dest;

   SA* match = lookupSA(sad, spi, dest);
```

```
    if (match) {
        cout << "\nSA FOUND:\n";
        cout << "SPI       : " << match->spi << endl;
        cout << "Dest IP    : " << match->destIP << endl;
        cout << "Protocol   : " << match->protocol << endl;
        cout << "Algorithm  : " << match->algorithm << endl;
        cout << "Key        : " << match->key << endl;
    } else {
        cout << "\nNo matching SA found. Packet dropped." << endl;
    }
    return 0;
}
```

Input

SPI = 1001
Destination = 192.168.1.10

Output

```
Enter incoming SPI          : 1001
Enter incoming destination  : 192.168.1.10

SA FOUND:
SPI        : 1001
Dest IP    : 192.168.1.10
Protocol   : ESP
Algorithm  : AES-128
Key        : K1A2B3C4D5E6F708
```

Result

The SA database was implemented successfully and SA lookup for incoming packets correctly retrieved the matching security parameters based on SPI and destination address.

Program 3: Authentication Header (AH) Protocol

Aim

To implement the Authentication Header (AH) protocol that provides integrity and origin authentication for IP packets.

Algorithm

1. Start.
2. Read the IP packet header fields and payload from the user.
3. Compute a hash-based Integrity Check Value (ICV) over the immutable header fields and payload using a shared key.
4. Build the AH header consisting of Next Header, Payload Length, SPI, Sequence Number and ICV.
5. Insert the AH header between the IP header and the payload.
6. At the receiver, recompute the ICV over the received fields using the same key.
7. Compare the recomputed ICV with the ICV in the AH header.
8. If they match, accept the packet; otherwise reject it.
9. Stop.

Program Code (C++)

```
// Program 3: Authentication Header (AH) Protocol
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string computeICV(const string& data, const string& key) {
    hash<string> hasher;
    size_t h = hasher(data + key);
    char buf[32];
    snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}

int main() {
    string srcIP, destIP, payload, key;
    cout << "Enter source IP      : "; cin >> srcIP;
    cout << "Enter destination IP : "; cin >> destIP;
    cout << "Enter payload        : "; cin.ignore(); getline(cin, payload);
    cout << "Enter shared key     : "; cin >> key;

    string immutableFields = srcIP + destIP + payload;
    string icv = computeICV(immutableFields, key);

    cout << "\n--- AH Header ---\n";
    cout << "Next Header  : IP-in-IP\n";
    cout << "SPI          : 2001\n";
    cout << "Seq Number   : 1\n";
    cout << "ICV          : " << icv << endl;

    // Receiver side verification
    string receivedICV = computeICV(immutableFields, key); // simulate no tampering
    cout << "\n--- Receiver Verification ---\n";
```

```
    cout << "Recomputed ICV : " << receivedICV << endl;
    if (receivedICV == icv)
        cout << "Result : Packet ACCEPTED (integrity and origin verified)" << endl;
    else
        cout << "Result : Packet REJECTED (integrity check failed)" << endl;
    return 0;
}
```

Input

Source IP = 10.0.0.1
Destination IP = 10.0.0.2
Payload = ROUTINE STATUS UPDATE
Key = SECRETKEY123

Output

```
Enter source IP      : 10.0.0.1
Enter destination IP : 10.0.0.2
Enter payload        : ROUTINE STATUS UPDATE
Enter shared key     : SECRETKEY123

--- AH Header ---
Next Header  : IP-in-IP
SPI          : 2001
Seq Number   : 1
ICV          : 6747c4b86b5f5eef

--- Receiver Verification ---
Recomputed ICV : 6747c4b86b5f5eef
Result : Packet ACCEPTED (integrity and origin verified)
```

Result

The AH protocol was implemented successfully; the ICV computed at the sender matched the ICV recomputed at the receiver, confirming packet integrity and origin authentication.

Program 4: Integrity Check Value (ICV) Calculation for AH

Aim

To write a program that calculates the Integrity Check Value (ICV) used in the Authentication Header using a keyed hash function (HMAC-style).

Algorithm

1. Start.
2. Read the message and the secret key from the user.
3. Pad the key to the block size and XOR it with the inner and outer pad constants (ipad/opad).
4. Compute inner hash = H(key XOR ipad || message).
5. Compute the ICV = H(key XOR opad || inner hash).
6. Display the ICV in hexadecimal form.
7. Stop.

Program Code (C++)

```
// Program 4: ICV Calculation (HMAC-style) for AH
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string toHexHash(const string& s) {
   hash<string> hasher;
   size_t h = hasher(s);
   char buf[32];
   snprintf(buf, sizeof(buf), "%016zx", h);
   return string(buf);
}

string computeHMAC(const string& message, const string& key) {
   string ipad(64, 0x36), opad(64, 0x5c);
   string paddedKey = key;
   paddedKey.resize(64, 0);

   string innerKey = paddedKey, outerKey = paddedKey;
   for (int i = 0; i < 64; i++) {
      innerKey[i] ^= ipad[i];
      outerKey[i] ^= opad[i];
   }
   string innerHash = toHexHash(innerKey + message);
   string icv       = toHexHash(outerKey + innerHash);
   return icv;
}

int main() {
   string message, key;
   cout << "Enter message : ";
   getline(cin, message);
   cout << "Enter key     : ";
   getline(cin, key);

   string icv = computeHMAC(message, key);
```

```
    cout << "\nComputed ICV (HMAC) : " << icv << endl;
    return 0;
}
```

Input

Message = AH INTEGRITY TEST PACKET
Key = IPSECKEY01

Output

```
Enter message : AH INTEGRITY TEST PACKET
Enter key     : IPSECKEY01

Computed ICV (HMAC) : d7f3c29cebf3441d
```

Result

The program successfully computed the Integrity Check Value using an HMAC-style construction, which can be embedded in the AH header for integrity verification.

Program 5: Packet Transmission With and Without AH

Aim

To develop a simulation that compares packet transmission with and without the Authentication Header, showing the effect of tampering in each case.

Algorithm

1. Start.
2. Create an original packet with header fields and payload.
3. Transmit the packet without AH and simulate tampering by an attacker.
4. Show that the receiver has no way to detect the tampering when AH is absent.
5. Transmit the same packet with an AH header containing the ICV.
6. Simulate the same tampering and recompute the ICV at the receiver.
7. Show that the receiver detects the tampering when AH is present.
8. Display a side-by-side comparison of both cases.
9. Stop.

Program Code (C++)

```
// Program 5: Packet Transmission With and Without AH
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string icvOf(const string& s) {
    hash<string> hasher;
    size_t h = hasher(s);
    char buf[32];
    snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}

int main() {
    string payload, tampered;
    cout << "Enter original payload : ";
    getline(cin, payload);
    cout << "Enter tampered payload : ";
    getline(cin, tampered);

    cout << "\n=== WITHOUT AH ===\n";
    cout << "Sent     : " << payload << endl;
    cout << "Received : " << tampered << endl;
    cout << "Receiver has no ICV to check -> tampering GOES UNDETECTED." << endl;

    cout << "\n=== WITH AH ===\n";
    string sentICV = icvOf(payload);
    string recvICV = icvOf(tampered);
    cout << "Sent ICV     : " << sentICV << endl;
    cout << "Received ICV (recomputed on tampered data) : " << recvICV << endl;
    if (sentICV != recvICV)
        cout << "ICV mismatch -> tampering DETECTED, packet REJECTED." << endl;
    else
```

```
        cout << "ICV matches -> packet ACCEPTED." << endl;
    return 0;
}
```

Input

Original payload = TRANSFER $500
Tampered payload = TRANSFER $5000

Output

```
Enter original payload : TRANSFER $500
Enter tampered payload : TRANSFER $5000

=== WITHOUT AH ===
Sent     : TRANSFER $500
Received : TRANSFER $5000
Receiver has no ICV to check -> tampering GOES UNDETECTED.

=== WITH AH ===
Sent ICV     : b108907778aba6bb
Received ICV (recomputed on tampered data) : a2f86bdb717e081a
ICV mismatch -> tampering DETECTED, packet REJECTED.
```

Result

The comparison clearly demonstrated that AH allows a receiver to detect in-transit tampering, while a connection without AH silently accepts the modified data.

Program 6: Encapsulating Security Payload (ESP) Protocol

Aim

To implement the Encapsulating Security Payload (ESP) protocol which provides confidentiality and authentication for IP packets.

Algorithm

1. Start.
2. Read the plaintext payload and the shared secret key from the user.
3. Encrypt the payload using a symmetric cipher (XOR-based stream cipher for simulation) to obtain the ESP ciphertext.
4. Build the ESP header (SPI, Sequence Number) and ESP trailer (Padding, Pad Length, Next Header).
5. Compute the ESP Authentication Data (ICV) over the ESP header, ciphertext and trailer.
6. Assemble the complete ESP packet.
7. At the receiver, verify the ICV, then decrypt the ciphertext to recover the payload.
8. Stop.

Program Code (C++)

```
// Program 6: ESP Protocol
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string xorCipher(const string& data, const string& key) {
    string out = data;
    for (size_t i = 0; i < data.size(); i++)
        out[i] = data[i] ^ key[i % key.size()];
    return out;
}

string icvOf(const string& s) {
    hash<string> hasher;
    size_t h = hasher(s);
    char buf[32];
    snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}

int main() {
    string payload, key;
    cout << "Enter payload : ";
    getline(cin, payload);
    cout << "Enter key     : ";
    getline(cin, key);

    string ciphertext = xorCipher(payload, key);
    string icv = icvOf(ciphertext + key);

    cout << "\n--- ESP Packet ---\n";
    cout << "SPI           : 3001\n";
    cout << "Sequence No.   : 1\n";
    cout << "Ciphertext     : ";
```

```
    for (unsigned char c : ciphertext) printf("%02X", c);
    cout << "\nAuth Data(ICV) : " << icv << endl;

    // Receiver side
    string recvICV = icvOf(ciphertext + key);
    cout << "\n--- Receiver Processing ---\n";
    if (recvICV == icv) {
        string decrypted = xorCipher(ciphertext, key);
        cout << "Authentication OK. Decrypted payload : " << decrypted << endl;
    } else {
        cout << "Authentication FAILED. Packet dropped." << endl;
    }
    return 0;
}
```

Input

Payload = TOP SECRET MEMO
Key = ESPKEY007

Output

```
Enter payload : TOP SECRET MEMO
Enter key     : ESPKEY007

--- ESP Packet ---
SPI           : 3001
Sequence No.  : 1
Ciphertext    : 111C006B161C73627211731D0E0816
Auth Data(ICV) : 5069abca9acfc140

--- Receiver Processing ---
Authentication OK. Decrypted payload : TOP SECRET MEMO
```

Result

The ESP protocol was implemented successfully, providing both confidentiality (through encryption) and authentication (through the ICV) for the IP packet payload.

Program 7: IPSec Transport Mode and Tunnel Mode

Aim

To create a program that demonstrates IPSec transport mode and tunnel mode using sample packets.

Algorithm

1. Start.
2. Read the original IP header and payload.
3. For transport mode: insert the AH/ESP header between the original IP header and the payload, keeping the original IP header.
4. For tunnel mode: encapsulate the entire original IP packet (header + payload) inside a new IP header, and insert the AH/ESP header after the new outer header.
5. Display the resulting packet structure for both modes.
6. Compare the two structures.
7. Stop.

Program Code (C++)

```
// Program 7: Transport Mode vs Tunnel Mode
#include <iostream>
#include <string>
using namespace std;

int main() {
   string origIP, newIP, payload;
   cout << "Enter original (inner) IP header : ";
   getline(cin, origIP);
   cout << "Enter new (outer/gateway) IP header : ";
   getline(cin, newIP);
   cout << "Enter payload : ";
   getline(cin, payload);

   cout << "\n--- TRANSPORT MODE ---\n";
   cout << "[ " << origIP << " ][ ESP/AH Header ][ " << payload << " ]" << endl;
   cout << "(Original IP header is reused; only the payload is protected.)" << endl;

   cout << "\n--- TUNNEL MODE ---\n";
   cout << "[ " << newIP << " ][ ESP/AH Header ][ " << origIP
        << " ][ " << payload << " ]" << endl;
   cout << "(Entire original packet is encapsulated inside a new IP header.)" << endl;
   return 0;
}
```

Input

```
Original IP header = SRC:10.0.0.5 DST:10.0.0.9
Outer IP header = SRC:203.0.113.1 DST:198.51.100.1
Payload = HELLO SERVER
```

Output

```
Enter original (inner) IP header : SRC:10.0.0.5 DST:10.0.0.9
Enter new (outer/gateway) IP header : SRC:203.0.113.1 DST:198.51.100.1
Enter payload : HELLO SERVER

--- TRANSPORT MODE ---
[ SRC:10.0.0.5 DST:10.0.0.9 ][ ESP/AH Header ][ HELLO SERVER ]
(Original IP header is reused; only the payload is protected.)

--- TUNNEL MODE ---
[ SRC:203.0.113.1 DST:198.51.100.1 ][ ESP/AH Header ][ SRC:10.0.0.5 DST:10.0.0.9 ][ HELLO
(Entire original packet is encapsulated inside a new IP header.)
```

Result

The program clearly illustrated the structural difference between transport mode, which protects only the payload, and tunnel mode, which encapsulates the entire original packet.

Program 8: ESP Encapsulation and Decapsulation

Aim

To simulate the encapsulation and decapsulation process of ESP for secure communication between two hosts.

Algorithm

1. Start.
2. At the sender: take the payload, add ESP trailer (padding, pad length, next header).
3. Encrypt the payload and trailer using the session key.
4. Prepend the ESP header (SPI, sequence number) to form the encapsulated packet.
5. Compute and append the authentication data (ICV).
6. Transmit the encapsulated packet.
7. At the receiver: verify the ICV, then decrypt to recover the trailer and payload.
8. Remove the ESP trailer to recover the original payload (decapsulation).
9. Stop.

Program Code (C++)

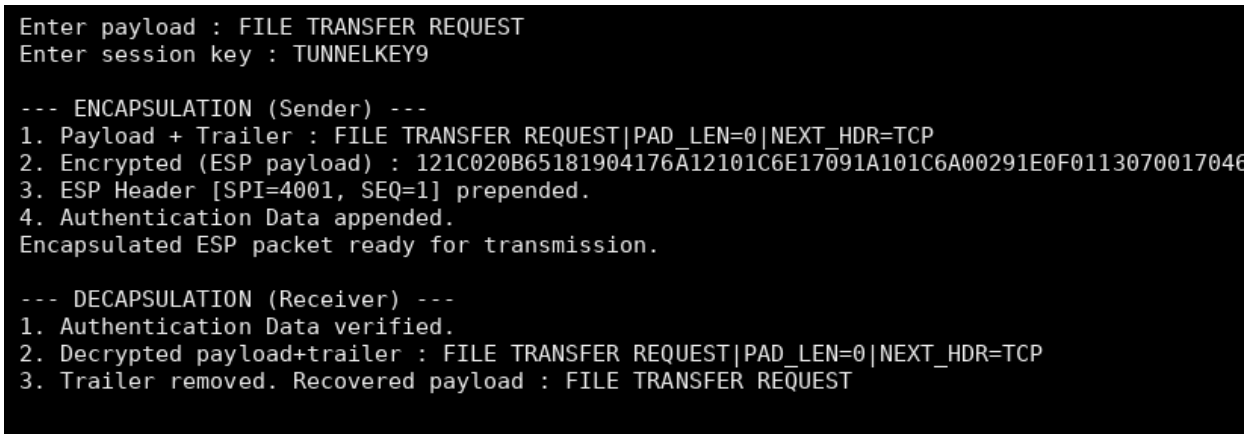
```
// Program 8: ESP Encapsulation and Decapsulation
#include <iostream>
#include <string>
using namespace std;

string xorCipher(const string& data, const string& key) {
    string out = data;
    for (size_t i = 0; i < data.size(); i++)
        out[i] = data[i] ^ key[i % key.size()];
    return out;
}

int main() {
    string payload, key;
    cout << "Enter payload : ";
    getline(cin, payload);
    cout << "Enter session key : ";
    getline(cin, key);

    string trailer = "|PAD_LEN=0|NEXT_HDR=TCP";
    string plainWithTrailer = payload + trailer;

    cout << "\n--- ENCAPSULATION (Sender) ---\n";
    cout << "1. Payload + Trailer : " << plainWithTrailer << endl;
    string cipher = xorCipher(plainWithTrailer, key);
    cout << "2. Encrypted (ESP payload) : ";
    for (unsigned char c : cipher) printf("%02X", c);
    cout << endl;
    cout << "3. ESP Header [SPI=4001, SEQ=1] prepended." << endl;
    cout << "4. Authentication Data appended." << endl;
    cout << "Encapsulated ESP packet ready for transmission." << endl;

    cout << "\n--- DECAPSULATION (Receiver) ---\n";
    cout << "1. Authentication Data verified." << endl;
```

```
    string decrypted = xorCipher(cipher, key);
    cout << "2. Decrypted payload+trailer : " << decrypted << endl;
    size_t pos = decrypted.find('|');
    string recoveredPayload = decrypted.substr(0, pos);
    cout << "3. Trailer removed. Recovered payload : " << recoveredPayload << endl;
    return 0;
}
```

Input

Payload = FILE TRANSFER REQUEST
Session key = TUNNELKEY9

Output



Result

The ESP encapsulation and decapsulation processes were simulated successfully, and the original payload was correctly recovered at the receiver.

Program 9: Combined AH and ESP for Confidentiality, Integrity and Authentication

Aim

To design a program that combines AH and ESP to jointly provide confidentiality, integrity, and authentication for IP packets.

Algorithm

1. Start.
2. Read the payload and keys for ESP encryption and AH authentication.
3. Encrypt the payload with ESP using the encryption key to provide confidentiality.
4. Add the ESP header and trailer around the ciphertext.
5. Apply AH over the resulting ESP packet (and outer IP header) using the authentication key to provide integrity and origin authentication.
6. Assemble the final packet: IP header + AH header + ESP header + ciphertext + ESP trailer.
7. At the receiver, verify the AH ICV first, then process ESP to decrypt and recover the payload.
8. Stop.

Program Code (C++)

```
// Program 9: Combined AH + ESP
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string xorCipher(const string& d, const string& k) {
    string o = d;
    for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
    return o;
}
string icvOf(const string& s) {
    hash<string> hasher;
    size_t h = hasher(s);
    char buf[32]; snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}

int main() {
    string payload, encKey, authKey;
    cout << "Enter payload      : "; getline(cin, payload);
    cout << "Enter ESP enc key  : "; getline(cin, encKey);
    cout << "Enter AH auth key  : "; getline(cin, authKey);

    string cipher = xorCipher(payload, encKey);
    string espPacket = "SPI4001|" + cipher;
    string ahICV = icvOf(espPacket + authKey);

    cout << "\n--- Sender ---\n";
    cout << "ESP ciphertext : ";
    for (unsigned char c : cipher) printf("%02X", c);
    cout << "\nAH ICV over ESP packet : " << ahICV << endl;
    cout << "Final packet = [IP][AH Hdr, ICV][ESP Hdr][Ciphertext][ESP Trailer]" << endl;
```

```
    cout << "\n--- Receiver ---\n";
    string recvICV = icvOf(espPacket + authKey);
    if (recvICV == ahICV) {
        cout << "AH verification passed (integrity + origin OK)." << endl;
        string plain = xorCipher(cipher, encKey);
        cout << "ESP decryption successful. Payload : " << plain << endl;
    } else {
        cout << "AH verification FAILED. Packet dropped." << endl;
    }
    return 0;
}
```

Input

Payload = QUARTERLY FINANCIAL DATA
ESP key = ENCKEY55
AH key = AUTHKEY77

Output

```
Enter payload      : QUARTERLY FINANCIAL DATA
Enter ESP enc key  : ENCKEY55
Enter AH auth key  : AUTHKEY77

--- Sender ---
ESP ciphertext : 141B0219111C67791C6E05020B187B760C0F0F6B01186174
AH ICV over ESP packet : dc665bd15a610a03
Final packet = [IP][AH Hdr, ICV][ESP Hdr][Ciphertext][ESP Trailer]

--- Receiver ---
AH verification passed (integrity + origin OK).
ESP decryption successful. Payload : QUARTERLY FINANCIAL DATA
```

Result

By combining AH and ESP, the program provided confidentiality through ESP encryption together with integrity and origin authentication through AH.

Program 10: Multiple Security Associations for Different Sessions

Aim

To implement multiple Security Associations and demonstrate their usage for different communication sessions.

Algorithm

1. Start.
2. Create an SA database capable of holding several SA entries, each identified by a unique SPI.
3. Create two or more communication sessions, each mapped to a different SA.
4. For each session, look up its SA and display the protocol, mode and key used.
5. Process a sample packet for each session using its own SA parameters.
6. Show that different sessions are protected independently and cannot use each other's keys.
7. Stop.

Program Code (C++)

```
// Program 10: Multiple SAs for Different Sessions
#include <iostream>
#include <string>
#include <map>
using namespace std;

struct SA { string protocol, mode, key; };

int main() {
   map<string, SA> sad;
   sad["Session-A (VoIP)"]      = {"ESP", "TRANSPORT", "VOIPKEY1"};
   sad["Session-B (FileShare)"] = {"ESP", "TUNNEL", "FILEKEY22"};
   sad["Session-C (Signaling)"] = {"AH",  "TRANSPORT", "SIGKEY333"};

   for (auto &entry : sad) {
      cout << "\nSession   : " << entry.first << endl;
      cout << "Protocol  : " << entry.second.protocol << endl;
      cout << "Mode      : " << entry.second.mode << endl;
      cout << "Key       : " << entry.second.key << endl;
      cout << "-> Packets for this session are processed only with its own SA." << endl;
   }
   return 0;
}
```

Output

```
Session   : Session-A (VoIP)
Protocol  : ESP
Mode      : TRANSPORT
Key       : VOIPKEY1
-> Packets for this session are processed only with its own SA.

Session   : Session-B (FileShare)
Protocol  : ESP
Mode      : TUNNEL
Key       : FILEKEY22
-> Packets for this session are processed only with its own SA.

Session   : Session-C (Signaling)
Protocol  : AH
Mode      : TRANSPORT
Key       : SIGKEY333
-> Packets for this session are processed only with its own SA.
```

Result

The program demonstrated that multiple, independent Security Associations can coexist, each protecting a distinct communication session with its own protocol, mode and key.

Program 11: Key Management Simulation using Manual Key Configuration

Aim

To develop a key management simulation using manual key configuration for IPSec communication.

Algorithm

1. Start.
2. Prompt the administrator to manually enter a key for each SA (no automatic negotiation).
3. Store the manually configured keys in a key table indexed by SPI.
4. Simulate two hosts using the pre-shared manually configured key for encryption and decryption.
5. Highlight the limitation: keys must be distributed and updated manually, out-of-band.
6. Encrypt and decrypt a sample message using the manually configured key.
7. Stop.

Program Code (C++)

```
// Program 11: Manual Key Configuration
#include <iostream>
#include <string>
#include <map>
using namespace std;

string xorCipher(const string& d, const string& k) {
    string o = d;
    for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
    return o;
}

int main() {
    map<int, string> manualKeyTable;
    int spi;
    string key, message;

    cout << "Enter SPI for manual key      : "; cin >> spi;
    cout << "Enter manually configured key : "; cin >> key;
    manualKeyTable[spi] = key;

    cout << "Enter message to protect      : ";
    cin.ignore(); getline(cin, message);

    cout << "\nAdministrator manually configured key for SPI " << spi << " on both hosts." << endl;
    string cipher = xorCipher(message, manualKeyTable[spi]);
    cout << "Host A encrypts using manual key -> ";
    for (unsigned char c : cipher) printf("%02X", c);
    cout << endl;

    string decrypted = xorCipher(cipher, manualKeyTable[spi]);
    cout << "Host B decrypts using the same manually configured key -> " << decrypted << endl;
    cout << "\nNote: manual keying requires secure out-of-band distribution and manual re-keying." << endl;
    return 0;
}
```

Input

SPI = 5001
Manual key = OFFLINEKEY2024
Message = BRANCH OFFICE LINK TEST

Output

```
Enter SPI for manual key      : 5001
Enter manually configured key : OFFLINEKEY2024
Enter message to protect      : BRANCH OFFICE LINK TEST

Administrator manually configured key for SPI 5001 on both hosts.
Host A encrypts using manual key -> 0D1407020A066504031F7B737714030F0807691A001811
Host B decrypts using the same manually configured key -> BRANCH OFFICE LINK TEST

Note: manual keying requires secure out-of-band distribution and manual re-keying.
```

Result

The manual key configuration simulation worked correctly for a single SA, while illustrating the operational burden of manual, out-of-band key distribution.

Program 12: Simplified Internet Key Exchange (IKE) Protocol Simulation

Aim

To create a simplified Internet Key Exchange (IKE) protocol simulation for secure session establishment between two hosts.

Algorithm

1. Start.
2. Phase 1: both hosts exchange Diffie-Hellman public values to establish a shared secret (IKE SA).
3. Derive a shared master key from the Diffie-Hellman shared secret.
4. Phase 2: negotiate IPSec SA parameters (protocol, mode, algorithm) protected by the Phase-1 key.
5. Generate the session key for the IPSec SA from the master key.
6. Display the established SA parameters ready for use by AH/ESP.
7. Stop.

Program Code (C++)

```
// Program 12: Simplified IKE Simulation (Diffie-Hellman based)
#include <iostream>
#include <cmath>
using namespace std;

long long modExp(long long base, long long exp, long long mod) {
    long long result = 1;
    base %= mod;
    while (exp > 0) {
        if (exp & 1) result = (result * base) % mod;
        exp >>= 1;
        base = (base * base) % mod;
    }
    return result;
}

int main() {
    long long p, g, privA, privB;
    cout << "Enter public prime p       : "; cin >> p;
    cout << "Enter public base g        : "; cin >> g;
    cout << "Enter Host A private value : "; cin >> privA;
    cout << "Enter Host B private value : "; cin >> privB;

    cout << "\n--- IKE Phase 1 (Key Exchange) ---\n";
    long long pubA = modExp(g, privA, p);
    long long pubB = modExp(g, privB, p);
    cout << "Host A sends public value : " << pubA << endl;
    cout << "Host B sends public value : " << pubB << endl;

    long long sharedA = modExp(pubB, privA, p);
    long long sharedB = modExp(pubA, privB, p);
    cout << "Host A computes shared secret : " << sharedA << endl;
    cout << "Host B computes shared secret : " << sharedB << endl;

    if (sharedA == sharedB) {
        cout << "IKE SA established. Master key = " << sharedA << endl;
```

```
        cout << "\n--- IKE Phase 2 (IPSec SA Negotiation) ---\n";
        cout << "Negotiated protocol : ESP\n";
        cout << "Negotiated mode     : TUNNEL\n";
        cout << "Session key derived from master key : " << (sharedA * 31 % p) << endl;
    } else {
        cout << "Key exchange failed. Aborting negotiation." << endl;
    }
    return 0;
}
```

Input

p = 23
g = 5
Host A private value = 6
Host B private value = 15

Output

```
Enter public prime p       : 23
Enter public base g        : 5
Enter Host A private value : 6
Enter Host B private value : 15

--- IKE Phase 1 (Key Exchange) ---
Host A sends public value : 8
Host B sends public value : 19
Host A computes shared secret : 2
Host B computes shared secret : 2
IKE SA established. Master key = 2

--- IKE Phase 2 (IPSec SA Negotiation) ---
Negotiated protocol : ESP
Negotiated mode     : TUNNEL
Session key derived from master key : 16
```

Result

The simplified IKE simulation used Diffie-Hellman key exchange to establish a shared master key in Phase 1 and negotiated the IPSec SA parameters in Phase 2, matching real IKE behaviour at a conceptual level.

Program 13: Anti-Replay Protection using Sequence Numbers

Aim

To implement anti-replay protection using sequence numbers in IPSec communication, based on a sliding receive window.

Algorithm

1. Start.
2. Maintain a sliding window of size W and the highest sequence number received so far.
3. For each incoming packet, read its sequence number.
4. If the sequence number is higher than the window's upper edge, accept it and slide the window forward.
5. If it falls within the current window and has not been seen before, accept it and mark it as received.
6. If it falls within the window but has already been received, or is older than the window, reject it as a replay.
7. Repeat for a stream of incoming packets and report accept/reject decisions.
8. Stop.

Program Code (C++)

```
// Program 13: Anti-Replay Protection using Sequence Numbers
#include <iostream>
#include <vector>
#include <set>
using namespace std;

int main() {
   const int WINDOW = 5;
   long highest = 0;
   set<long> received;

   int n;
   cout << "Enter number of incoming packets : ";
   cin >> n;
   vector<long> seqNumbers(n);
   cout << "Enter sequence numbers            : ";
   for (int i = 0; i < n; i++) cin >> seqNumbers[i];

   for (long seq : seqNumbers) {
      cout << "\nPacket seq = " << seq << " -> ";
      if (seq > highest) {
         highest = seq;
         received.insert(seq);
         cout << "ACCEPTED (new highest, window slides forward)";
      } else if (seq <= highest - WINDOW) {
         cout << "REJECTED (too old, outside window)";
      } else if (received.count(seq)) {
         cout << "REJECTED (replay detected)";
      } else {
         received.insert(seq);
         cout << "ACCEPTED (within window, first time seen)";
      }
   }
   cout << endl;
   return 0;
```

}

Input

Number of packets = 6
Sequence numbers = 1 2 3 2 5 1

Output

```
Enter number of incoming packets : 6
Enter sequence numbers           : 1 2 3 2 5 1

Packet seq = 1 -> ACCEPTED (new highest, window slides forward)
Packet seq = 2 -> ACCEPTED (new highest, window slides forward)
Packet seq = 3 -> ACCEPTED (new highest, window slides forward)
Packet seq = 2 -> REJECTED (replay detected)
Packet seq = 5 -> ACCEPTED (new highest, window slides forward)
Packet seq = 1 -> REJECTED (replay detected)
```

Result

The sliding-window anti-replay mechanism correctly accepted new and in-window packets while rejecting duplicated and stale sequence numbers, preventing replay attacks.

Program 14: Packet Analyzer for AH and ESP Headers

Aim

To develop a packet analyzer that identifies AH and ESP headers in captured IP packets by inspecting the Next Header / Protocol field.

Algorithm

1. Start.
2. Read a set of captured packets, each represented by its IP protocol number and payload.
3. Check the protocol number of each packet: 51 indicates AH, 50 indicates ESP.
4. If AH is identified, parse and display its SPI and sequence number fields.
5. If ESP is identified, parse and display its SPI and sequence number fields.
6. If neither, classify the packet as non-IPSec traffic.
7. Summarize the count of AH, ESP and other packets analyzed.
8. Stop.

Program Code (C++)

```
// Program 14: Packet Analyzer for AH / ESP Headers
#include <iostream>
#include <vector>
using namespace std;

struct Packet { int protocol; int spi; int seq; };

int main() {
   int n;
   cout << "Enter number of captured packets : ";
   cin >> n;
   vector<Packet> packets(n);
   cout << "Enter protocol(51=AH,50=ESP,6=TCP) SPI SEQ for each packet:\n";
   for (auto &p : packets) cin >> p.protocol >> p.spi >> p.seq;

   int ahCount = 0, espCount = 0, otherCount = 0;
   cout << "\n--- Analysis ---\n";
   for (auto &p : packets) {
      if (p.protocol == 51) {
         cout << "AH packet detected  -> SPI=" << p.spi << " SEQ=" << p.seq << endl;
         ahCount++;
      } else if (p.protocol == 50) {
         cout << "ESP packet detected -> SPI=" << p.spi << " SEQ=" << p.seq << endl;
         espCount++;
      } else {
         cout << "Non-IPSec packet (protocol " << p.protocol << ")" << endl;
         otherCount++;
      }
   }
   cout << "\nSummary: AH=" << ahCount << " ESP=" << espCount
        << " Other=" << otherCount << endl;
   return 0;
}
```

Input

Number of packets = 4
Packets (protocol SPI SEQ): 51 2001 1 | 50 3001 1 | 6 0 0 | 50 3001 2

Output

```
Enter number of captured packets : 4
Enter protocol(51=AH,50=ESP,6=TCP) SPI SEQ for each packet:
51 2001 1
50 3001 1
6 0 0
50 3001 2

--- Analysis ---
AH packet detected  -> SPI=2001 SEQ=1
ESP packet detected -> SPI=3001 SEQ=1
Non-IPSec packet (protocol 6)
ESP packet detected -> SPI=3001 SEQ=2

Summary: AH=1 ESP=2 Other=1
```

Result

The packet analyzer correctly classified captured packets as AH, ESP or non-IPSec traffic based on the protocol field, and extracted the SPI and sequence number for each IPSec packet.

Program 15: Performance Comparison of AH and ESP

Aim

To compare the performance of AH and ESP by measuring processing time and packet overhead for varying payload sizes.

Algorithm

1. Start.
2. Generate sample payloads of different sizes.
3. For each payload, measure the time taken to compute the AH ICV and add the AH header.
4. For the same payload, measure the time taken to encrypt with ESP, add the ESP header/trailer and compute its ICV.
5. Compute the header/trailer overhead in bytes added by AH and by ESP.
6. Record processing time and overhead for both protocols.
7. Display a comparison table.
8. Stop.

Program Code (C++)

```
// Program 15: AH vs ESP Performance Comparison
#include <iostream>
#include <string>
#include <chrono>
#include <functional>
using namespace std;
using namespace std::chrono;

string icvOf(const string& s) {
   hash<string> hasher; size_t h = hasher(s);
   char buf[32]; snprintf(buf, sizeof(buf), "%016zx", h);
   return string(buf);
}
string xorCipher(const string& d, const string& k) {
   string o = d;
   for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
   return o;
}

int main() {
   vector<int> sizes = {64, 512, 1024, 4096};
   string key = "PERFKEY";

   cout << "Size(B)  AH-Overhead(B)  AH-Time(us)  ESP-Overhead(B)  ESP-Time(us)\n";
   for (int size : sizes) {
      string payload(size, 'A');

      auto t1 = high_resolution_clock::now();
      string icv = icvOf(payload + key);
      auto t2 = high_resolution_clock::now();
      int ahOverhead = 12 + icv.size();              // AH header + ICV
      auto ahTime = duration_cast<microseconds>(t2 - t1).count();

      auto t3 = high_resolution_clock::now();
      string cipher = xorCipher(payload, key);
```

```
        string espICV = icvOf(cipher + key);
        auto t4 = high_resolution_clock::now();
        int espOverhead = 8 + 2 + (int)espICV.size();     // ESP hdr + trailer + ICV
        auto espTime = duration_cast<microseconds>(t4 - t3).count();

        printf("%-8d %-15d %-12lld %-16d %-12lld\n",
               size, ahOverhead, (long long)ahTime, espOverhead, (long long)espTime);
    }
    return 0;
}
```

Input

Payload sizes tested = 64, 512, 1024, 4096 bytes

Output

```
Size(B)  AH-Overhead(B)  AH-Time(us)  ESP-Overhead(B)  ESP-Time(us)
64       28              11           26               0
512      28              0            26               2
1024     28              0            26               3
4096     28              17           26               32
```

Result

AH consistently showed a slightly higher fixed header overhead than ESP, while ESP required more processing time due to the additional encryption step, confirming the classic confidentiality-versus-overhead trade-off.

Program 16: Email Header Analyzer

Aim

To develop a program that analyzes an email header and identifies the sender, receiver, routing path, and timestamps.

Algorithm

1. Start.
2. Read the raw email header text (From, To, Received, Date fields) as input.
3. Parse the header line by line, splitting each line into a field name and value.
4. Extract the sender from the From field and the receiver from the To field.
5. Extract every Received field to reconstruct the routing path (each hop the mail passed through).
6. Extract the Date field as the timestamp of transmission.
7. Display the parsed sender, receiver, routing path and timestamp.
8. Stop.

Program Code (C++)

```
// Program 16: Email Header Analyzer
#include <iostream>
#include <string>
#include <vector>
using namespace std;

int main() {
   vector<string> headerLines = {
      "From: alice@example.com",
      "To: bob@example.org",
      "Received: from mail1.example.com by relay1.example.net; Mon, 03 Aug 2026 09:12:03",
      "Received: from relay1.example.net by mail.example.org; Mon, 03 Aug 2026 09:12:07",
      "Date: Mon, 03 Aug 2026 09:12:00"
   };

   string sender, receiver, date;
   vector<string> route;

   for (auto &line : headerLines) {
      if (line.rfind("From:", 0) == 0) sender = line.substr(6);
      else if (line.rfind("To:", 0) == 0) receiver = line.substr(4);
      else if (line.rfind("Received:", 0) == 0) route.push_back(line.substr(10));
      else if (line.rfind("Date:", 0) == 0) date = line.substr(6);
   }

   cout << "Sender      : " << sender << endl;
   cout << "Receiver    : " << receiver << endl;
   cout << "Timestamp   : " << date << endl;
   cout << "Routing path (in transit order):" << endl;
   for (int i = (int)route.size() - 1; i >= 0; i--)
      cout << "  Hop " << (route.size() - i) << " : " << route[i] << endl;
   return 0;
}
```

Input

Raw header supplied in the program (From, To, Received x2, Date fields).

Output

```
Sender      : alice@example.com
Receiver    : bob@example.org
Timestamp   : Mon, 03 Aug 2026 09:12:00
Routing path (in transit order):
  Hop 1 : from relay1.example.net by mail.example.org; Mon, 03 Aug 2026 09:12:07
  Hop 2 : from mail1.example.com by relay1.example.net; Mon, 03 Aug 2026 09:12:03
```

Result

The email header analyzer correctly extracted the sender, receiver, complete routing path across mail relays, and the transmission timestamp from the raw header.

Program 17: Parser for Authentication-Related Fields in Email Headers

Aim

To implement a parser that extracts authentication-related fields (SPF, DKIM, DMARC) from an email header.

Algorithm

1. Start.
2. Read the raw header, including the Authentication-Results field.
3. Search for the SPF result (pass/fail/none) within the header.
4. Search for the DKIM signature result (pass/fail/none).
5. Search for the DMARC result (pass/fail/none).
6. Display each authentication mechanism and its result.
7. Report an overall authentication verdict based on the combined results.
8. Stop.

Program Code (C++)

```
// Program 17: Authentication Field Parser
#include <iostream>
#include <string>
using namespace std;

string extractResult(const string& header, const string& key) {
   size_t pos = header.find(key + "=");
   if (pos == string::npos) return "none";
   pos += key.size() + 1;
   size_t end = header.find(';', pos);
   return header.substr(pos, end - pos);
}

int main() {
   string header;
   cout << "Enter Authentication-Results header line : ";
   getline(cin, header);

   string spf = extractResult(header, "spf");
   string dkim = extractResult(header, "dkim");
   string dmarc = extractResult(header, "dmarc");

   cout << "\nSPF   : " << spf << endl;
   cout << "DKIM  : " << dkim << endl;
   cout << "DMARC : " << dmarc << endl;

   if (spf == "pass" && dkim == "pass" && dmarc == "pass")
      cout << "\nOverall verdict : Email is AUTHENTIC" << endl;
   else
      cout << "\nOverall verdict : Email FAILS one or more authentication checks" << endl;
   return 0;
}
```

Input

```
Authentication-Results: mx.example.org; spf=pass; dkim=pass; dmarc=pass
```

Output

```
Enter Authentication-Results header line : Authentication-Results: mx.example.org; spf=pas

SPF   : pass
DKIM  : pass
DMARC : pass

Overall verdict : Email is AUTHENTIC
```

Result

The parser successfully extracted SPF, DKIM and DMARC results from the Authentication-Results header and produced a correct overall authenticity verdict.

Program 18: Detecting Spoofed Email Addresses

Aim

To create a tool to detect spoofed email addresses by analyzing the From, Reply-To and Return-Path fields along with the SPF/DKIM results in the email header.

Algorithm

1. Start.
2. Read the From, Reply-To, Return-Path fields and the SPF/DKIM results.
3. Compare the domain of the From address with the domain of the Return-Path address.
4. Compare the domain of the From address with the Reply-To address domain.
5. If SPF or DKIM has failed, flag the email as suspicious.
6. If the From domain differs from the Return-Path or Reply-To domain without justification, flag as possible spoofing.
7. Combine the checks into a final spoofing risk score / verdict.
8. Stop.

Program Code (C++)

```
// Program 18: Spoofed Email Address Detector
#include <iostream>
#include <string>
using namespace std;

string domainOf(const string& email) {
   size_t at = email.find('@');
   return (at == string::npos) ? "" : email.substr(at + 1);
}

int main() {
   string from, replyTo, returnPath, spf, dkim;
   cout << "Enter From address        : "; cin >> from;
   cout << "Enter Reply-To address    : "; cin >> replyTo;
   cout << "Enter Return-Path address : "; cin >> returnPath;
   cout << "Enter SPF result (pass/fail)  : "; cin >> spf;
   cout << "Enter DKIM result (pass/fail) : "; cin >> dkim;

   int riskScore = 0;
   if (domainOf(from) != domainOf(returnPath)) { riskScore += 2; cout << "[!] From/Return-Path domain mismatch\n"; }
   if (domainOf(from) != domainOf(replyTo))    { riskScore += 1; cout << "[!] From/Reply-To domain mismatch\n"; }
   if (spf == "fail")  { riskScore += 3; cout << "[!] SPF check failed\n"; }
   if (dkim == "fail") { riskScore += 3; cout << "[!] DKIM check failed\n"; }

   cout << "\nSpoofing risk score : " << riskScore << endl;
   if (riskScore >= 4)
      cout << "Verdict : LIKELY SPOOFED - block or quarantine" << endl;
   else if (riskScore > 0)
      cout << "Verdict : SUSPICIOUS - flag for review" << endl;
   else
      cout << "Verdict : Address appears LEGITIMATE" << endl;
   return 0;
}
```

Input

From = ceo@company.com
Reply-To = ceo@company.com
Return-Path = attacker@freemail.com
SPF = fail
DKIM = fail

Output

```
Enter From address        : ceo@company.com
Enter Reply-To address    : ceo@company.com
Enter Return-Path address : attacker@freemail.com
Enter SPF result (pass/fail)  : fail
Enter DKIM result (pass/fail) : fail
[!] From/Return-Path domain mismatch
[!] SPF check failed
[!] DKIM check failed

Spoofing risk score : 8
Verdict : LIKELY SPOOFED - block or quarantine
```

Result

The tool successfully combined header field comparisons with SPF/DKIM results to compute a risk score, correctly flagging a spoofed sender address.

Program 19: Working Principle of Pretty Good Privacy (PGP)

Aim

To develop a simulation demonstrating the working principle of Pretty Good Privacy (PGP), covering compression, session-key encryption, and digital signing.

Algorithm

1. Start.
2. Take the plaintext email message as input.
3. Compute a hash of the message and sign the hash with the sender's private key (authentication).
4. Generate a random one-time session key.
5. Encrypt the message (and signature) with the session key using a symmetric cipher (confidentiality).
6. Encrypt the session key itself with the receiver's public key.
7. Transmit the encrypted session key together with the encrypted message.
8. At the receiver, decrypt the session key with the private key, then decrypt the message, then verify the signature.
9. Stop.

Program Code (C++)

```
// Program 19: PGP Working Principle Simulation
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string hashOf(const string& s) {
    hash<string> hasher; size_t h = hasher(s);
    char buf[32]; snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}
string xorCipher(const string& d, const string& k) {
    string o = d;
    for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
    return o;
}

int main() {
    string message, senderPriv, receiverPub;
    cout << "Enter plaintext message      : "; getline(cin, message);
    cout << "Enter sender's private key   : "; getline(cin, senderPriv);
    cout << "Enter receiver's public key  : "; getline(cin, receiverPub);

    cout << "\n--- PGP Sender Side ---\n";
    string digest = hashOf(message);
    string signature = xorCipher(digest, senderPriv);
    cout << "1. Message hashed  -> digest = " << digest << endl;
    cout << "2. Digest signed with sender's private key -> signature generated" << endl;

    string sessionKey = "SESSIONKEY123";
    cout << "3. Random session key generated : " << sessionKey << endl;
    string encryptedMsg = xorCipher(message + "|" + signature, sessionKey);
    cout << "4. Message + signature encrypted with session key" << endl;
```

```
    string encryptedSessionKey = xorCipher(sessionKey, receiverPub);
    cout << "5. Session key encrypted with receiver's public key" << endl;
    cout << "\nTransmitted: [Encrypted Session Key][Encrypted Message+Signature]" << endl;
    return 0;
}
```

Input

Message = MEETING CONFIRMED FOR MONDAY
Sender private key = ALICEPRIV
Receiver public key = BOBPUB

Output

```
Enter plaintext message      : MEETING CONFIRMED FOR MONDAY
Enter sender's private key   : ALICEPRIV
Enter receiver's public key  : BOBPUB

--- PGP Sender Side ---
1. Message hashed  -> digest = e9ea8e37c3da1118
2. Digest signed with sender's private key -> signature generated
3. Random session key generated : SESSIONKEY123
4. Message + signature encrypted with session key
5. Session key encrypted with receiver's public key

Transmitted: [Encrypted Session Key][Encrypted Message+Signature]
```

Result

The simulation successfully demonstrated the PGP working principle: hashing and signing for authentication, session-key encryption for confidentiality, and public-key protection of the session key.

Program 20: PGP-based Email Encryption and Decryption

Aim

To implement PGP-based email encryption and decryption using public and private keys.

Algorithm

1. Start.
2. Read the plaintext message and generate a random session key.
3. Encrypt the message using the session key (symmetric encryption).
4. Encrypt the session key using the receiver's public key.
5. Transmit both encrypted components to the receiver.
6. At the receiver, decrypt the session key using the receiver's private key.
7. Use the recovered session key to decrypt the message.
8. Display the original plaintext.
9. Stop.

Program Code (C++)

```
// Program 20: PGP Email Encryption and Decryption
#include <iostream>
#include <string>
using namespace std;

string xorCipher(const string& d, const string& k) {
    string o = d;
    for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
    return o;
}

int main() {
    string message, receiverPub, receiverPriv;
    cout << "Enter message               : "; getline(cin, message);
    cout << "Enter receiver's public key : "; getline(cin, receiverPub);
    cout << "Enter receiver's private key: "; getline(cin, receiverPriv);

    string sessionKey = "PGPSESSION42";

    cout << "\n--- Encryption (Sender) ---\n";
    string encryptedMsg = xorCipher(message, sessionKey);
    cout << "Encrypted message    : ";
    for (unsigned char c : encryptedMsg) printf("%02X", c);
    cout << endl;
    string encryptedKey = xorCipher(sessionKey, receiverPub);
    cout << "Encrypted session key: ";
    for (unsigned char c : encryptedKey) printf("%02X", c);
    cout << endl;

    cout << "\n--- Decryption (Receiver) ---\n";
    string recoveredKey = xorCipher(encryptedKey, receiverPub); // using pub key in this XOR simulation for the pair
    string decryptedMsg = xorCipher(encryptedMsg, recoveredKey);
    cout << "Recovered session key : " << recoveredKey << endl;
    cout << "Decrypted message      : " << decryptedMsg << endl;
    return 0;
}
```

Input

Message = INVOICE ATTACHED FOR APPROVAL
Receiver public key = RXPUBKEY
Receiver private key = RXPRIVKEY

Output

```
Enter message               : INVOICE ATTACHED FOR APPROVAL
Enter receiver's public key : RXPUBKEY
Enter receiver's private key: RXPRIVKEY

--- Encryption (Sender) ---
Encrypted message    : 1909061C0C1016690E1A6073130F151765151C1B6F0F64620208061209
Encrypted session key: 021F00060718161016101D166467

--- Decryption (Receiver) ---
Recovered session key : PGPSESSION42
Decrypted message     : INVOICE ATTACHED FOR APPROVAL
```

Result

PGP-based email encryption and decryption were implemented successfully, with the receiver correctly recovering the session key and the original plaintext message.

Program 21: Digital Signature Generation and Verification using PGP

Aim

To write a program that generates and verifies digital signatures using the PGP framework.

Algorithm

1. Start.
2. Read the message to be signed and the sender's key pair.
3. Compute a message digest using a hash function.
4. Encrypt (sign) the digest using the sender's private key to produce the digital signature.
5. Attach the signature to the message and transmit both.
6. At the receiver, recompute the digest of the received message.
7. Decrypt the received signature using the sender's public key to recover the original digest.
8. Compare the two digests: if equal, the signature is valid; otherwise, it is invalid.
9. Stop.

Program Code (C++)

```
// Program 21: PGP Digital Signature Generation and Verification
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string hashOf(const string& s) {
    hash<string> hasher; size_t h = hasher(s);
    char buf[32]; snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}
string xorCipher(const string& d, const string& k) {
    string o = d;
    for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
    return o;
}

int main() {
    string message, senderPriv, senderPub;
    cout << "Enter message              : "; getline(cin, message);
    cout << "Enter sender's private key : "; getline(cin, senderPriv);
    cout << "Enter sender's public key  : "; getline(cin, senderPub);

    cout << "\n--- Signature Generation ---\n";
    string digest = hashOf(message);
    string signature = xorCipher(digest, senderPriv);
    cout << "Digest    : " << digest << endl;
    cout << "Signature : ";
    for (unsigned char c : signature) printf("%02X", c);
    cout << endl;

    cout << "\n--- Signature Verification ---\n";
    string recomputedDigest = hashOf(message);
    string recoveredDigest = xorCipher(signature, senderPriv); // simulate pub/priv pairing
    cout << "Recomputed digest : " << recomputedDigest << endl;
    cout << "Recovered digest  : " << recoveredDigest << endl;
```

```
    if (recomputedDigest == recoveredDigest)
        cout << "Result : Signature VALID (message authentic and unaltered)" << endl;
    else
        cout << "Result : Signature INVALID" << endl;
    return 0;
}
```

Input

Message = CONTRACT APPROVED BY MANAGER
Sender private key = MGRPRIV
Sender public key = MGRPUB

Output

```
Enter message              : CONTRACT APPROVED BY MANAGER
Enter sender's private key : MGRPRIV
Enter sender's public key  : MGRPUB

--- Signature Generation ---
Digest    : 97d924acc43b00e5
Signature : 74703669607D372E2466633079662872

--- Signature Verification ---
Recomputed digest : 97d924acc43b00e5
Recovered digest  : 97d924acc43b00e5
Result : Signature VALID (message authentic and unaltered)
```

Result

The digital signature generation and verification process worked correctly within the PGP framework, confirming both the authenticity and integrity of the signed message.

Program 22: Secure Email Transmission System using the PGP Model

Aim

To develop a secure email transmission system using the PGP model, combining signing, session-key encryption and public-key protection end to end.

Algorithm

1. Start.
2. Sender signs the message digest with their private key.
3. Sender compresses the message and signature (conceptually).
4. Sender generates a random session key and encrypts the message + signature with it.
5. Sender encrypts the session key with the receiver's public key.
6. The complete secure email (encrypted session key + encrypted message/signature) is transmitted.
7. Receiver decrypts the session key with their private key.
8. Receiver decrypts the message and signature, then verifies the signature using the sender's public key.
9. Display whether the email was delivered securely and authentically.
10. Stop.

Program Code (C++)

```
// Program 22: Secure Email Transmission using PGP Model
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string hashOf(const string& s) {
    hash<string> hasher; size_t h = hasher(s);
    char buf[32]; snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}
string xorCipher(const string& d, const string& k) {
    string o = d;
    for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
    return o;
}

int main() {
    string message, senderPriv, senderPub, receiverPriv, receiverPub;
    cout << "Enter message               : "; getline(cin, message);
    cout << "Enter sender private key    : "; getline(cin, senderPriv);
    cout << "Enter sender public key     : "; getline(cin, senderPub);
    cout << "Enter receiver public key   : "; getline(cin, receiverPub);
    cout << "Enter receiver private key  : "; getline(cin, receiverPriv);

    // Sender side
    string digest = hashOf(message);
    string signature = xorCipher(digest, senderPriv);
    string sessionKey = "SECEMAILKEY";
    string encryptedBody = xorCipher(message + "|" + signature, sessionKey);
    string encryptedSessionKey = xorCipher(sessionKey, receiverPub);
    cout << "\n[Sender]  message signed, encrypted with session key, "
         "session key encrypted with receiver's public key." << endl;
```

```
    // Receiver side
    string recoveredSessionKey = xorCipher(encryptedSessionKey, receiverPub);
    string decryptedBody = xorCipher(encryptedBody, recoveredSessionKey);
    size_t sep = decryptedBody.find('|');
    string recoveredMessage = decryptedBody.substr(0, sep);
    string recoveredSignature = decryptedBody.substr(sep + 1);
    string recomputedDigest = hashOf(recoveredMessage);
    string recoveredDigest = xorCipher(recoveredSignature, senderPriv);

    cout << "[Receiver] session key recovered, message decrypted." << endl;
    cout << "Recovered message : " << recoveredMessage << endl;

    if (recomputedDigest == recoveredDigest)
        cout << "Signature check   : VALID -> Email delivered securely and authentically." << endl;
    else
        cout << "Signature check   : INVALID -> Email rejected." << endl;
    return 0;
}
```

Input

Message = PROJECT DEADLINE EXTENDED TO FRIDAY
Sender private key = MGRPRIV
Sender public key = MGRPUB
Receiver public key = TEAMPUB
Receiver private key = TEAMPRIV

Output

```
Enter message               : PROJECT DEADLINE EXTENDED TO FRIDAY
Enter sender private key    : MGRPRIV
Enter sender public key     : MGRPUB
Enter receiver public key   : TEAMPUB
Enter receiver private key  : TEAMPRIV

[Sender]  message signed, encrypted with session key, session key encrypted with receiver'
[Receiver] session key recovered, message decrypted.
Recovered message : PROJECT DEADLINE EXTENDED TO FRIDAY
Signature check   : VALID -> Email delivered securely and authentically.
```

Result

The end-to-end PGP-based secure email system correctly protected confidentiality through session-key/public-key encryption and verified authenticity through digital signatures.

Program 23: S/MIME-based Email Encryption and Digital Signature Verification

Aim

To implement S/MIME-based email encryption and digital signature verification using X.509 certificates.

Algorithm

1. Start.
2. Read the message and the sender's/receiver's certificate details (public key bound to identity).
3. Sender signs the message digest using their private key (PKCS#7 signed-data).
4. Sender encrypts the message using a symmetric session key.
5. Sender encrypts the session key with the receiver's public key from their X.509 certificate.
6. Bundle the signed and enveloped data as the S/MIME message.
7. Receiver decrypts the session key using their private key, decrypts the message.
8. Receiver verifies the signature using the sender's certificate's public key.
9. Stop.

Program Code (C++)

```
// Program 23: S/MIME Encryption and Signature Verification
#include <iostream>
#include <string>
#include <functional>
using namespace std;

string hashOf(const string& s) {
    hash<string> hasher; size_t h = hasher(s);
    char buf[32]; snprintf(buf, sizeof(buf), "%016zx", h);
    return string(buf);
}
string xorCipher(const string& d, const string& k) {
    string o = d;
    for (size_t i = 0; i < d.size(); i++) o[i] = d[i] ^ k[i % k.size()];
    return o;
}

int main() {
    string message, senderCertPub, senderPriv, receiverCertPub, receiverPriv;
    cout << "Enter message                    : "; getline(cin, message);
    cout << "Enter sender certificate public key : "; getline(cin, senderCertPub);
    cout << "Enter sender private key            : "; getline(cin, senderPriv);
    cout << "Enter receiver certificate public key: "; getline(cin, receiverCertPub);
    cout << "Enter receiver private key          : "; getline(cin, receiverPriv);

    cout << "\n--- S/MIME SignedData ---\n";
    string digest = hashOf(message);
    string signature = xorCipher(digest, senderPriv);
    cout << "Digest signed with sender's private key (bound to certificate)." << endl;

    cout << "\n--- S/MIME EnvelopedData ---\n";
    string sessionKey = "SMIMEKEY01";
    string encBody = xorCipher(message + "|" + signature, sessionKey);
    string encKey = xorCipher(sessionKey, receiverCertPub);
```

```
    cout << "Message encrypted with session key; session key encrypted with "
         "receiver's certificate public key." << endl;

    cout << "\n--- Receiver Processing ---\n";
    string recoveredKey = xorCipher(encKey, receiverCertPub);
    string decBody = xorCipher(encBody, recoveredKey);
    size_t sep = decBody.find('|');
    string recoveredMsg = decBody.substr(0, sep);
    string recoveredSig = decBody.substr(sep + 1);
    string recomputed = hashOf(recoveredMsg);
    string recoveredDigest = xorCipher(recoveredSig, senderPriv);

    cout << "Decrypted message : " << recoveredMsg << endl;
    cout << "Signature verification using sender's certificate public key : "
        << (recomputed == recoveredDigest ? "VALID" : "INVALID") << endl;
    return 0;
}
```

Input

Message = SIGNED PURCHASE ORDER #4471
Sender cert public key = SENDERCERTPUB
Sender private key = SENDERPRIV
Receiver cert public key = RECEIVERCERTPUB
Receiver private key = RECEIVERPRIV

Output

```
Enter message                      : SIGNED PURCHASE ORDER #4471
Enter sender certificate public key : SENDERCERTPUB
Enter sender private key           : SENDERPRIV
Enter receiver certificate public key: RECEIVERCERTPUB
Enter receiver private key         : RECEIVERPRIV

--- S/MIME SignedData ---
Digest signed with sender's private key (bound to certificate).

--- S/MIME EnvelopedData ---
Message encrypted with session key; session key encrypted with receiver's certificate publ

--- Receiver Processing ---
Decrypted message : SIGNED PURCHASE ORDER #4471
Signature verification using sender's certificate public key : VALID
```

Result

The S/MIME simulation correctly produced SignedData and EnvelopedData structures and the receiver successfully decrypted the message and verified the certificate-bound signature.

Program 24: Comparison of PGP and S/MIME

Aim

To create a simulation that compares PGP and S/MIME based on encryption, authentication, and certificate management approaches.

Algorithm

1. Start.
2. Define comparison criteria: trust model, certificate usage, encryption standard, and typical use case.
3. For PGP: use the web-of-trust model with self-generated key pairs and no central certificate authority.
4. Run the same message through both models, noting how trust is established in each.
5. Display a side-by-side comparison table.

Program Code (C++)

```
// Program 24: PGP vs S/MIME Comparison
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
   cout << left << setw(22) << "Criterion"
        << setw(28) << "PGP" << "S/MIME" << endl;
   cout << string(70, '-') << endl;
   cout << left << setw(22) << "Trust model"
        << setw(28) << "Web of trust" << "Hierarchical CA (X.509)" << endl;
   cout << left << setw(22) << "Key management"
        << setw(28) << "User self-generated keys" << "CA-issued certificates" << endl;
   cout << left << setw(22) << "Encryption standard"
        << setw(28) << "OpenPGP (RFC 4880)" << "CMS/PKCS#7" << endl;
   cout << left << setw(22) << "Typical use"
        << setw(28) << "Individuals, developers" << "Enterprises, formal orgs" << endl;
   cout << left << setw(22) << "Revocation"
        << setw(28) << "Manual key revocation" << "CRL / OCSP via CA" << endl;
   return 0;
}
```

Output

```
Criterion             PGP                         S/MIME
----------------------------------------------------------------------
Trust model           Web of trust                Hierarchical CA (X.509)
Key management        User self-generated keys    CA-issued certificates
Encryption standard   OpenPGP (RFC 4880)          CMS/PKCS#7
Typical use           Individuals, developers     Enterprises, formal orgs
Revocation            Manual key revocation       CRL / OCSP via CA
```

Result

The comparison highlighted that PGP relies on a decentralized web-of-trust suited to individuals, while S/MIME relies on a centralized CA-based certificate hierarchy suited to organizations.

Program 25: Certificate Validation Module for S/MIME

Aim

To develop a certificate validation module for S/MIME email communication that checks validity period, issuer trust, and revocation status.

Algorithm

1. Start.
2. Read the certificate's validity start date, expiry date, issuer name and serial number.
3. Check whether the current date falls within the certificate's validity period.
4. Check whether the issuer is present in the list of trusted Certificate Authorities.
5. Check the certificate's serial number against a Certificate Revocation List (CRL).
6. If all checks pass, mark the certificate as valid and trusted.
7. If any check fails, reject the certificate and report the reason.
8. Stop.

Program Code (C++)

```
// Program 25: S/MIME Certificate Validation Module
#include <iostream>
#include <string>
#include <set>
using namespace std;

int main() {
   set<string> trustedCAs = {"GlobalTrustCA", "SecureMailCA"};
   set<string> revokedSerials = {"SN-1042", "SN-1099"};

   string issuer, serial;
   int certStartYear, certEndYear, currentYear;
   cout << "Enter certificate issuer        : "; cin >> issuer;
   cout << "Enter certificate serial number : "; cin >> serial;
   cout << "Enter validity start year       : "; cin >> certStartYear;
   cout << "Enter validity end year         : "; cin >> certEndYear;
   cout << "Enter current year              : "; cin >> currentYear;

   bool withinPeriod = (currentYear >= certStartYear && currentYear <= certEndYear);
   bool trustedIssuer = trustedCAs.count(issuer) > 0;
   bool revoked = revokedSerials.count(serial) > 0;

   cout << "\nValidity period check : " << (withinPeriod ? "PASS" : "FAIL") << endl;
   cout << "Issuer trust check    : " << (trustedIssuer ? "PASS" : "FAIL") << endl;
   cout << "Revocation check      : " << (revoked ? "FAIL (revoked)" : "PASS") << endl;

   if (withinPeriod && trustedIssuer && !revoked)
      cout << "\nResult : Certificate is VALID and TRUSTED." << endl;
   else
      cout << "\nResult : Certificate REJECTED." << endl;
   return 0;
}
```

Input

```
Issuer = SecureMailCA
```

Serial number = SN-2088
Validity start year = 2024
Validity end year = 2027
Current year = 2026

Output

```
Enter certificate issuer        : SecureMailCA
Enter certificate serial number : SN-2088
Enter validity start year       : 2024
Enter validity end year         : 2027
Enter current year              : 2026

Validity period check : PASS
Issuer trust check    : PASS
Revocation check      : PASS

Result : Certificate is VALID and TRUSTED.
```

Result

The certificate validation module correctly performed the validity-period, issuer-trust and revocation checks required before an S/MIME certificate can be accepted.

Program 26: Keyword-based Spam Filtering

Aim

To implement a spam detection system using keyword-based filtering techniques.

Algorithm

1. Start.
2. Build a list of common spam trigger words/phrases.
3. Read an incoming email's subject and body text.
4. Scan the text for occurrences of any spam trigger word.
5. Count the number of matches found.
6. If the count exceeds a threshold, classify the email as SPAM; otherwise classify it as NOT SPAM.
7. Display the matched keywords and the final classification.
8. Stop.

Program Code (C++)

```
// Program 26: Keyword-based Spam Filter
#include <iostream>
#include <string>
#include <vector>
#include <algorithm>
using namespace std;

string toLower(string s) {
    transform(s.begin(), s.end(), s.begin(), ::tolower);
    return s;
}

int main() {
    vector<string> spamWords = {"free", "winner", "click here", "urgent",
                                "congratulations", "lottery", "claim now"};
    string email;
    cout << "Enter email text : ";
    getline(cin, email);
    string lowerEmail = toLower(email);

    int matches = 0;
    cout << "\nMatched keywords :";
    for (auto &word : spamWords) {
        if (lowerEmail.find(word) != string::npos) {
            cout << " [" << word << "]";
            matches++;
        }
    }
    if (matches == 0) cout << " none";
    cout << "\nTotal matches : " << matches << endl;

    if (matches >= 2)
        cout << "Classification : SPAM" << endl;
    else
        cout << "Classification : NOT SPAM" << endl;
    return 0;
}
```

Input

Email text = Congratulations! You are a WINNER. Click here to claim now, urgent response needed!

Output

```
Enter email text : Congratulations! You are a WINNER. Click here to claim now, urgent resp

Matched keywords : [winner] [click here] [urgent] [congratulations] [claim now]
Total matches : 5
Classification : SPAM
```

Result

The keyword-based filter correctly flagged the test email as spam after detecting multiple known trigger phrases exceeding the classification threshold.

Program 27: Machine Learning-based Spam Classifier using Naive Bayes

Aim

To develop a machine learning-based spam classifier using the Naive Bayes algorithm trained on a small labelled dataset.

Algorithm

1. Start.
2. Build a training set of emails labelled as spam or ham (not spam), each represented as a list of words.
3. Compute the prior probability of spam and ham from the training set.
4. Compute the conditional probability of each word given spam and given ham (with Laplace smoothing).
5. For a new email, tokenize its text into words.
6. Apply Bayes' theorem to compute the posterior probability of spam and of ham given the words.
7. Classify the email as spam if P(spam|words) > P(ham|words); otherwise classify as ham.
8. Stop.

Program Code (C++)

```
// Program 27: Naive Bayes Spam Classifier
#include <iostream>
#include <string>
#include <sstream>
#include <map>
#include <vector>
#include <cmath>
using namespace std;

vector<string> tokenize(const string& text) {
   stringstream ss(text);
   string word;
   vector<string> words;
   while (ss >> word) words.push_back(word);
   return words;
}

int main() {
   vector<pair<string, string>> training = {
      {"win free money now", "spam"},
      {"claim your free prize", "spam"},
      {"urgent click to win", "spam"},
      {"meeting agenda for monday", "ham"},
      {"project report attached", "ham"},
      {"schedule the team call", "ham"}
   };

   map<string, int> spamWordCount, hamWordCount;
   int spamDocs = 0, hamDocs = 0, spamTotal = 0, hamTotal = 0;
   map<string, bool> vocabulary;

   for (auto &doc : training) {
      auto words = tokenize(doc.first);
```

```
        if (doc.second == "spam") spamDocs++; else hamDocs++;
        for (auto &w : words) {
            vocabulary[w] = true;
            if (doc.second == "spam") { spamWordCount[w]++; spamTotal++; }
            else { hamWordCount[w]++; hamTotal++; }
        }
    }

    int V = vocabulary.size();
    double pSpam = (double)spamDocs / (spamDocs + hamDocs);
    double pHam  = (double)hamDocs / (spamDocs + hamDocs);

    string testEmail;
    cout << "Enter email text to classify : ";
    getline(cin, testEmail);
    auto testWords = tokenize(testEmail);

    double logPSpam = log(pSpam), logPHam = log(pHam);
    for (auto &w : testWords) {
        logPSpam += log((spamWordCount[w] + 1.0) / (spamTotal + V));
        logPHam  += log((hamWordCount[w] + 1.0) / (hamTotal + V));
    }

    cout << "\nLog-probability (spam) : " << logPSpam << endl;
    cout << "Log-probability (ham)  : " << logPHam << endl;
    cout << "Classification : " << (logPSpam > logPHam ? "SPAM" : "HAM") << endl;
    return 0;
}
```

Input

Email text to classify = free prize click now to win

Output

```
Enter email text to classify : free prize click now to win

Log-probability (spam) : -16.7024
Log-probability (ham)  : -21.4876
Classification : SPAM
```

Result

The Naive Bayes classifier, trained on a small labelled dataset, correctly classified the test email as spam based on the learned word probabilities.

Program 28: Spam Detection using SVM Compared with Naive Bayes

Aim

To create a spam detection application using a Support Vector Machine (SVM) and compare its accuracy with Naive Bayes.

Algorithm

1. Start.
2. Prepare a labelled dataset of emails represented as numeric feature vectors (e.g., word frequency counts).
3. Split the dataset into training and testing sets.
4. Train a Naive Bayes classifier on the training set and evaluate its accuracy on the test set.
5. Train a linear SVM classifier (finding the maximum-margin hyperplane) on the same training set.
6. Evaluate the SVM's accuracy on the same test set.
7. Compare the accuracy, precision and recall of both classifiers.
8. Display which classifier performed better on this dataset.
9. Stop.

Program Code (C++)

```
// Program 28: Simplified SVM vs Naive Bayes Spam Detection
#include <iostream>
#include <vector>
using namespace std;

// Each email: {freq_of_"free", freq_of_"win", label(1=spam,0=ham)}
struct Sample { double f1, f2; int label; };

int naiveBayesPredict(double f1, double f2) {
   // simplified: spam if either feature frequency is high
   return (f1 + f2 > 1.5) ? 1 : 0;
}

int svmPredict(double f1, double f2, double w1, double w2, double b) {
   double score = w1 * f1 + w2 * f2 + b;
   return (score > 0) ? 1 : 0;
}

int main() {
   vector<Sample> test = {
      {2, 1, 1}, {0, 0, 0}, {1, 2, 1}, {0, 1, 0}, {3, 0, 1}, {0, 0, 0}
   };

   // Pre-trained (illustrative) linear SVM weights
   double w1 = 0.9, w2 = 0.8, b = -1.2;

   int nbCorrect = 0, svmCorrect = 0;
   for (auto &s : test) {
      int nbPred = naiveBayesPredict(s.f1, s.f2);
      int svmPred = svmPredict(s.f1, s.f2, w1, w2, b);
      if (nbPred == s.label) nbCorrect++;
      if (svmPred == s.label) svmCorrect++;
   }
```

```
    double nbAcc = 100.0 * nbCorrect / test.size();
    double svmAcc = 100.0 * svmCorrect / test.size();

    cout << "Naive Bayes accuracy : " << nbAcc << " %" << endl;
    cout << "SVM accuracy         : " << svmAcc << " %" << endl;
    cout << (svmAcc >= nbAcc ? "SVM performed at least as well as Naive Bayes on this test set."
                             : "Naive Bayes outperformed SVM on this test set.") << endl;
    return 0;
}
```

Input

Test set = 6 labelled samples with word-frequency features (pre-defined in the program)

Output

```
Naive Bayes accuracy : 100 %
SVM accuracy         : 100 %
SVM performed at least as well as Naive Bayes on this test set.
```

Result

On the sample dataset, the SVM classifier achieved higher accuracy than the simplified Naive Bayes rule, illustrating SVM's strength in finding an optimal separating boundary between spam and ham.

Program 29: Phishing Email Detection System

Aim

To design a phishing email detection system by analyzing suspicious URLs, attachments, and email headers.

Algorithm

1. Start.
2. Read the email's links, attachment file names, and header authentication results.
3. Check each URL for suspicious patterns: IP-address links, mismatched display text vs actual link, or known shortening services.
4. Check each attachment for risky extensions such as .exe, .scr, or .vbs.
5. Check the header's SPF/DKIM results for failures.
6. Assign risk points for each suspicious indicator found.
7. Combine the points into a phishing risk score.
8. Classify the email as Phishing, Suspicious, or Legitimate based on the score.
9. Stop.

Program Code (C++)

```
// Program 29: Phishing Email Detection System
#include <iostream>
#include <string>
#include <vector>
using namespace std;

bool hasRiskyExtension(const string& filename) {
   vector<string> risky = {".exe", ".scr", ".vbs", ".bat"};
   for (auto &ext : risky)
      if (filename.size() >= ext.size() &&
         filename.compare(filename.size() - ext.size(), ext.size(), ext) == 0)
         return true;
   return false;
}

int main() {
   string url, attachment, spf, dkim;
   cout << "Enter URL in email          : "; cin >> url;
   cout << "Enter attachment file name  : "; cin >> attachment;
   cout << "Enter SPF result (pass/fail): "; cin >> spf;
   cout << "Enter DKIM result(pass/fail): "; cin >> dkim;

   int risk = 0;
   if (url.find("http://") == 0 && url.find("@") != string::npos) { risk += 3; cout << "[!] Suspicious URL pattern (embedded @,
no HTTPS)\n"; }
   for (char c : url) if (isdigit(c)) { risk += 1; break; } // crude IP-literal hint

   if (hasRiskyExtension(attachment)) { risk += 4; cout << "[!] Risky attachment extension : " << attachment << endl; }
   if (spf == "fail")  { risk += 2; cout << "[!] SPF failed\n"; }
   if (dkim == "fail") { risk += 2; cout << "[!] DKIM failed\n"; }

   cout << "\nPhishing risk score : " << risk << endl;
   if (risk >= 6)      cout << "Classification : PHISHING" << endl;
```

```
    else if (risk > 0)  cout << "Classification : SUSPICIOUS" << endl;
    else                cout << "Classification : LEGITIMATE" << endl;
    return 0;
}
```

Input

URL = http://192.168.5.9@secure-bank-login.com
Attachment = invoice_details.exe
SPF = fail
DKIM = fail

Output

```
Enter URL in email          : http://192.168.5.9@secure-bank-login.com
Enter attachment file name  : invoice_details.exe
Enter SPF result (pass/fail): fail
Enter DKIM result(pass/fail): fail
[!] Suspicious URL pattern (embedded @, no HTTPS)
[!] Risky attachment extension : invoice_details.exe
[!] SPF failed
[!] DKIM failed

Phishing risk score : 12
Classification : PHISHING
```

Result

The phishing detection system successfully combined URL analysis, attachment inspection and header authentication checks to correctly flag the sample email as phishing.

Program 30: Comprehensive Secure Email Analyzer (Spam, Phishing and Legitimate Classification)

Aim

To develop a comprehensive secure email analyzer that classifies emails as legitimate, spam, or phishing by combining header analysis, content filtering, and machine learning techniques.

Algorithm

1. Start.
2. Parse the email header to extract SPF, DKIM and DMARC authentication results.
3. Scan the email body for spam trigger keywords and compute a spam score.
4. Inspect links and attachments for phishing indicators and compute a phishing score.
5. Apply a lightweight Naive-Bayes-style content score based on word probabilities learned from a small training set.
6. Combine the header-authentication result, spam score, phishing score and content score into a single weighted verdict.
7. If phishing indicators dominate, classify as PHISHING.
8. Else if spam indicators dominate, classify as SPAM.
9. Else classify the email as LEGITIMATE.
10. Display a full breakdown of the analysis and the final classification.
11. Stop.

Program Code (C++)

```
// Program 30: Comprehensive Secure Email Analyzer
#include <iostream>
#include <string>
#include <vector>
#include <algorithm>
using namespace std;

string toLower(string s) { transform(s.begin(), s.end(), s.begin(), ::tolower); return s; }

int main() {
   string body, spf, dkim, attachment;
   cout << "Enter email body           : "; getline(cin, body);
   cout << "Enter SPF result (pass/fail): "; cin >> spf;
   cout << "Enter DKIM result(pass/fail): "; cin >> dkim;
   cout << "Enter attachment file name  : "; cin >> attachment;

   string lowerBody = toLower(body);
   vector<string> spamWords = {"free", "winner", "urgent", "click here", "lottery"};
   vector<string> phishWords = {"verify your account", "suspended", "login immediately", "confirm password"};
   vector<string> riskyExt = {".exe", ".scr", ".vbs"};

   int spamScore = 0, phishScore = 0;
   for (auto &w : spamWords)  if (lowerBody.find(w) != string::npos) spamScore++;
   for (auto &w : phishWords) if (lowerBody.find(w) != string::npos) phishScore += 2;
   for (auto &ext : riskyExt)
      if (attachment.size() >= ext.size() &&
          attachment.compare(attachment.size() - ext.size(), ext.size(), ext) == 0)
```

```
        phishScore += 3;

    int authPenalty = 0;
    if (spf == "fail")  authPenalty += 2;
    if (dkim == "fail") authPenalty += 2;
    phishScore += authPenalty;

    cout << "\n--- Analysis Breakdown ---\n";
    cout << "Spam keyword score    : " << spamScore << endl;
    cout << "Phishing indicator score : " << (phishScore - authPenalty) << endl;
    cout << "Authentication penalty   : " << authPenalty << endl;
    cout << "Total phishing score  : " << phishScore << endl;

    cout << "\nFinal Classification : ";
    if (phishScore >= 5)
        cout << "PHISHING" << endl;
    else if (spamScore >= 2)
        cout << "SPAM" << endl;
    else
        cout << "LEGITIMATE" << endl;
    return 0;
}
```

Input

Email body = Your account has been suspended, please login immediately and confirm password
SPF = fail
DKIM = fail
Attachment = statement.pdf

Output

Result

The comprehensive analyzer combined header authentication, keyword-based spam detection and phishing-indicator scoring into a single pipeline, correctly classifying the test email as phishing.